

is referred to additional information regarding specific exercises in the section regarding joint motion exercises. Strengthening exercises are used to improve the strength and speed of muscle contraction, whereas endurance exercises are performed to help with cardiovascular and musculoskeletal stamina. Although both types of conditioning may be performed, usually one form of conditioning is primarily undertaken, depending on the intended use of the dog. For example, a field trial dog will likely benefit more from endurance exercises, although some form of strengthening is also beneficial. Similarly, a racing greyhound may benefit more from strengthening exercises but endurance conditioning is also beneficial.

Strength exercises are those performed with maximal or near-maximal muscle contraction, with relatively few repetitions. The object of strength exercises is to increase the strength of muscle contraction. Myofiber and muscle hypertrophy occur with continued conditioning. However, there is little to no change in the aerobic capacity of the muscle.

Endurance exercises are those performed with repetitive contractions over a prolonged time, with relatively little load applied to the muscle. Although myofiber and muscle hypertrophy do not occur to any appreciable extent, the aerobic capacity and efficiency of muscle contraction are greatly enhanced.

Core Strengthening Exercises

The importance of strong core body muscles, including the back, abdominal, and upper limb muscles, has been recently emphasized in human rehabilitation and training. Strong core muscles make it easier to perform many physical activities and help provide support to the midsection when performing other exercises. Although these types of exercises are a bit more challenging in dogs because of the difficulty in consciously performing exercises that isolate particular muscle groups, there are several exercises that are useful and require relatively little training.

Caution should be used in performing any of these exercises in dogs with back, hip, or shoulder problems. In general, core-strength exercises may be repeated five times. As core strength improves, they can be increased to 10 to 15 repetitions, two to three times daily. A combination of exercises allows strengthening of different muscle groups, while preventing boredom.

Sit-to-Stands, Down-to-Stands on Ball, Physioroll or Trampoline

This variation of the sit-to-stand and down-to-stand is performed on an exercise ball, physio-roll, or trampoline. The additional challenge achieved by performing these activities on an unstable surface engages the core muscles, resulting in strengthening. These exercises are good to perform early in the course of a core strengthening program.

Standing on a Rolling Cart

The patient may be placed on a rolling cart, preferably with sides, and slowly pulled. Although this exercise also challenges proprioception, the core muscles are also engaged while trying to maintain balance. As the patient becomes stronger and corrects its position while being pulled in a straight line, turns and irregular movements may be incorporated during the therapy session for additional challenges and strengthening. The therapist should always provide support to prevent falls in the event that the patient loses its balance.

Rolling

Animals that roll from their side while in lateral recumbency or from their back in dorsal recumbency to a sternal position engage their abdominal muscles, in particular, their abdominal oblique muscles. If the dog can be trained to roll completely, that engages even more muscles and requires more strength. A low-calorie treat is helpful as a training aid.

Sit-Ups and Crunches and Hold, Twisting Crunches

Dogs may perform abdominal crunches or sit-ups with proper positioning and training. To perform this exercise, the handler sits on the floor with the legs spread far enough apart to support the dog in dorsal recumbency. A treat is then used to encourage the dog to lift its head and thorax up while maintaining the pelvis and lower limbs in contact with the ground and within the trough formed by the handler's legs. If possible, the treat may be held just out of the dog's reach to encourage the dog to hold the crunch for several seconds. In the beginning, it may be beneficial to have the dog receive a treat reward on every crunch. But as the training progresses, intermittent administration of a treat is likely adequate to keep the dog motivated. Peanut butter or Kong paste frozen into a Kong toy or coffee mug may work better than individual treats for some patients during this exercise because it is easier to get the patient to hold the desired position.

Another version of the abdominal crunch is the twisting crunch. To perform this exercise, the treat is held somewhat to the side, encouraging the dog to sit up in a diagonal fashion to reach the treat. This is one of the most effective crunch workouts, because it engages all of the abdominal muscle fibers at once.

Super Pointing

This exercise is relatively easy to perform and most dogs tolerate it well; however, two handlers may be necessary. The dog is placed in a square standing position and one handler maintains control of the dog's head and lifts one of the forelimbs. Another handler simultaneously lifts the diagonal rear limb. This position is held for several seconds. In patients with hindlimb injuries, it is often easier to lift the contralateral hindlimb first followed by the ipsilateral

forelimb. If possible, lifting of the legs may be done with an elastic resistance band so that the dog not only engages the core muscles, but also contracts the limb muscles to help maintain balance. Although having the handler lift both legs and hold them without bands also engages the core muscles, additional challenges and strengthening are achieved by using resistance bands. Another variation is to train the patient to shake or lift a forelimb while standing squarely without shifting weight significantly onto the hindlimbs or the contralateral side.

An easier version of this exercise is simply one-legged pointing. If only one handler is available, or if the dog is not strong enough or has proprioceptive deficits, one leg may be lifted and held at a time. The forelimbs and rear limbs should be raised one at a time to engage the different core muscles.

Hup and Praying Positions

Core strengthening may be achieved by having the dog stand with the forelimbs supported while the rear limbs remain on the ground, a maneuver sometimes called *hup* or a *standing plank position*. The handler may support the forelimbs of the dog or the limbs may be supported by a piece of furniture or stairs. This exercise is one of the earliest to help strengthen the core muscles. A reverse standing plank may be performed by placing the rear limbs in an elevated position while allowing the forelimbs to remain on the ground in a weight-bearing position.

Conversely, another maneuver is a praying stance. Some dogs do this naturally, whereas others will need some training to perform it correctly. With this position, the hindlimbs maintain a standing position while the forelimbs remain on the ground. It is important that the elbows be in contact with the ground to perform the exercise correctly. Clicker training is especially helpful to “capture” this position during training.

Plank on a Ball

A plank that provides additional challenges is one performed on a therapy ball or roll. The plank is performed by having the dog stand with its rear limbs on the floor and the forelimbs supported by a therapy ball or roll. This plank position may be held for several seconds. Additional challenges may be instituted by rolling the ball back and forth (ball roll-ins and roll-outs), side to side, or 360 degrees, doing ball circles. Initially, the excursions are kept small, but as strengthening and compliance improve, larger excursions may be instituted. For dogs requiring more support and less challenge, a therapy roll may be used that moves in only two dimensions. Alternatively, the rear limbs may be placed on the ball or roll, with the forelimbs on the ground. Again, additional challenges may be instituted by rolling the ball back and forth (ball roll-ins and roll-outs).

Another variation is for the dog to stand with its forelimbs on a donut, and use that as a pivot point around

which the rear limbs move laterally. In addition to normal core strengthening, this helps to strengthen the abdominal oblique muscles as well as the thigh abductor and adductor muscles. For symmetrical strengthening, the dog should move equal distances in both directions. A balance dome (half ball–half flat surface) or a BOSU ball may also be used.

For more challenge, a modified plank on a ball may be performed. With the dog positioned as previously described, a forelimb or rear limb is raised off the floor and held for 3 to 5 seconds. The exercise is repeated holding up the other limbs. Super pointing may also be performed with the forelimbs on the ball by lifting diagonal limbs at the same time.

Back Extension

The muscles responsible for back extension may be strengthened by going from a sitting to a begging position or jumping up on the rear limbs and holding this position while pursuing a treat. As strength and coordination improve, the dog should maintain the two-legged standing position for several seconds; some dogs may be training to “walk” on two legs. Plank exercises also help to strengthen back extensor muscles.

Super Standing

Dogs may stand with all four limbs on a therapy roll. Although maintaining balance in a static position is highly challenging to the core muscles, moving the roll from side



Figure 30-13 Dogs may stand with all four limbs on a therapy roll. Although maintaining balance in a static position is highly challenging to the core muscles, moving the roll from side to side and rocking front to back provides additional strengthening.



Figure 30-14 Standing on a balance dome (half ball, half flat surface) or a BOSU ball may be used for core strengthening exercises.

to side and rocking front to back provides additional strengthening (Figure 30-13). One of the most challenging core exercises is for the dog to place both forelimbs on a therapy ball or roll, and the rear limbs on a separate ball or roll. Initially, larger balls with more stability are easier for the dog. As core strengthening improves, smaller balls or rolls may be used. Alternatively, the dog may stand on inflated discs with the forelimbs on one disc and the rear limbs on a separate disc. For additional challenges, the balls, rolls, or disks may be rocked back and forth, and in the case of rolls and balls, they may be moved closer or farther apart from each other to engage different core muscles.

The use of paw pods is less challenging for those dogs beginning a training program. A balance dome (half ball, half flat surface) or a BOSU ball may also be used (Figure 30-14).

Incline Walking, Trotting, and Running

Walking or jogging the dog up or down inclines aids in strengthening of the quadriceps, semitendinosus, semimembranosus, gastrocnemius, and gluteal muscles while ascending, and the triceps and shoulder stabilizing muscles while descending a ramp. Muscle strength in the hips and stifles is required for the dog to propel itself up an incline, whereas muscle strength in the shoulders and elbows is required to brake while going down a declined surface. Incline walking may be used as a low-impact activity.

Exercise Considerations

Inclines should begin with gradual slopes and increase to progressively steeper slopes if possible. As the dog's



Figure 30-15 Additional strengthening may be provided by walking up an inclined surface while wearing a weighted vest or leg weights.

strength and stamina improve, the dog may be challenged by walking up longer, steeper slopes and by increasing the duration and speed of the climbing exercise. Further challenges are provided by doing inclines while wearing a weighted vest or leg weights (Figure 30-15). Jogging and running up slopes promotes strength and power.

Dancing and Wheelbarrowing

Dancing and wheelbarrowing enhances strength of the extensor muscles of the rear limbs and forelimbs, respectively. The amount of peak force placed on the limbs while dancing is slightly greater than what would occur at a trot, whereas wheelbarrowing results in peak force placed on the limb that is between walking and trotting.⁶ Therefore if the stage of recovery is not adequate to withstand these stresses, the exercises should not be undertaken until adequate healing has occurred.

Exercise Considerations

Consider using a muzzle on dogs when first starting these exercises. To perform dancing, the forelimbs are lifted off the ground, allowing the patient to bear weight only on the hindlimbs. The handler then moves the dog backward and forward. Dogs may be elevated as high as possible and also dance up and down inclines or hills to place additional stress on the hindlimbs. Furthermore, leg weights may be placed on the limbs for additional strengthening.

To perform the wheelbarrow exercise, the handler places the hands under the caudal abdomen and lifts the rear limbs of the dog off the ground, and the dog is moved forward. Dogs may be wheelbarrowed up and down inclines for greater muscle strengthening. Theoretically, the higher the animal is lifted, the greater the forces placed on the forelimbs. However, the increase in weight bearing may not be as great as anticipated because the stride length is much shorter when the animal is wheelbarrowed from a

greater height. Similar to dancing, leg weights may be placed on the limbs for additional strengthening.

Sit-to-Stand Exercises

Sit-to-stand exercises help strengthen hip and stifle extensor muscles. The act of sitting, then standing up, requires activation and strength of the gluteal, quadriceps, hamstring, and gastrocnemius muscle groups. The back and abdominal muscles are also affected to some extent.

Exercise Considerations

Some training may be necessary, and low-calorie treats may be offered as a training aid to provide motivation to perform the movement. It is important to perform these exercises correctly. Attention should be paid to sitting and standing straight, with no leaning to one side, and the joints of both rear limbs should be symmetrically flexed so that the dog sits squarely on its haunches. The handler then asks the dog to stand, take a few steps forward, and then again sit. The sit-to-stands may be repeated a number of times before the dog is allowed to rest. It may be easier in some cases to back the dog into a corner, with the affected limb next to a wall so that the dog cannot slide the limb out while rising or sitting. Start with 5 to 10 repetitions once or twice daily, and work up to 15 repetitions three to four times daily.

Gluteal muscle atrophy may occur in dogs with hip arthritis. The sit-to-stand exercise allows active contraction of the gluteal muscles, but the hip joint is not generally extended to a point that causes pain because the hip only extends to a standing position. Therefore strengthening of the gluteal muscles can occur without the excessive pain encountered with hip extension. Each patient should be assessed to be certain that sit-to-stands are not too stressful or painful for that individual. If normal sit-to-stands are too painful, or the patient is too weak to rise from a sitting position, several modifications may be tried. Doing sit-to-stands with the pelvis in an elevated position may be helpful in patients that are too weak to rise from a complete sitting position. Dogs may sit with their pelvis on a stool, step, stack of books, curb, firm cushion, or the therapist's leg or knee, and rise from this elevated position (Figure 30-16). Many dogs will need guidance from the therapist's hands to learn to sit onto an object because it may feel strange for some patients. Alternatively, dogs may be assisted with a sling from a sitting to a standing position. Dogs may also do sit-to-stands on an incline, with the body facing downhill to make it easier to stand. Alternatively, dogs may do sit-to-stands on an incline with the body facing uphill or sideways to each direction to provide an additional challenge and greater strengthening. For this challenge, dogs may perform five repetitions in each direction twice daily. Further challenges may be provided by applying a weighted vest or backpack to encourage additional muscle strengthening. Super sits can be done using



Figure 30-16 Dogs may sit with their pelvis on a step and rise from this elevated position to modify the sit to stand exercise.

a donut or physioroll by asking the patient to sit and stand while keeping the forelimbs on the device.

Rhythmic Stabilization

Rhythmic stabilization is a technique used to strengthen postural muscles in dogs with significant weakness of these muscles. In particular, this exercise is valuable for strengthening the triceps and quadriceps muscle groups. It also helps to provide core strengthening and neuromuscular feedback to joints, ligaments, and muscle-tendon units.

Exercise Considerations

Rhythmic stabilization may be performed with the patient on a therapy ball or roll, trampoline, or inflated mattress. With the animal standing in a square position, the therapist places pressure over the pelvis or interscapular region and gently “bounces” the animal up and down. With downward pressure, the animal will begin to sink, and then reflexively will contract the extensor muscles to recover to a normal standing position. The bouncing motion should be relatively rapid, with only enough recovery time to regain the normal standing position. Performing the exercises with the animal standing on a compressible surface generally prevents a well-trained animal from going to a sit position.

Hand Shaking

In some situations, the flexors of the elbow joint may be weak following conditions such as partial brachial plexus injury or intervertebral disk herniation. Having the dog do shake hands exercises with the affected limb helps with functional strengthening.

Exercise Considerations

Most dogs have been trained to shake hands with their owners, so this is a relatively easy exercise. It is most



Figure 30-17 Greater strengthening may be achieved by placement of a leg weight while doing shake hands exercises.

beneficial to begin with the affected limb on the ground with the dog in a sitting position. The therapist asks for the dog's paw, trying to achieve as high a placement as possible to achieve the greatest elbow flexion possible. Shoulder extensor muscle strengthening may also occur with this exercise. Initially, three to five repetitions may be adequate. Increased repetitions are added as the dog is able. Greater strengthening may be achieved by placement of a leg weight or resistance band on the antebrachium (Figure 30-17). To reduce the moment arm, the weight may be placed proximal on the radius and ulna. As the dog gains strength, the weight may be moved further distally or more weight may be added.

Stairs

Climbing stairs is useful to improve power in the rear-limb extensors, whereas descending stairs helps strengthen the forelimb extensors. In many instances, negotiating stairs is an activity of daily living that may be performed in the home environment. Quadriceps and gluteal muscle groups are strengthened as the animal pushes off, extending both hips and knees while propelling the body up the steps. Conversely, the biceps brachii, triceps, and extensors of the carpus are strengthened by descending stairs.

Exercise Considerations

The dog must begin slowly climbing stairs to encourage proper use of the rear limbs, as opposed to simply carrying the limb, hopping with both rear limbs, or skipping up stairs. Encourage the dog to go slowly and deliberately, climbing the stairs in a reciprocal stepping gait. Stairs should be introduced slowly because this is a challenging exercise for both the musculoskeletal and cardiovascular systems and the animal may fatigue quickly. Initially, some dogs may require assistance from the handler. Begin with

1 to 3 flights of stairs for early strengthening, and gradually increase to 5 to 10 flights of stairs once or twice daily. As the patient improves, dogs may jog and trot up stairs, and eventually may run up stairs.

Cavaletti Rails

Cavaletti rails may be used to encourage muscle strengthening and coordination through a greater active ROM of certain joints, and lengthen strides in all limbs. Similar to the shake hands exercise, they may be used to help increase strength of forelimb flexor muscles in patients with weak joint flexion. Unlike the shake hands exercise, however, walking over cavaletti rails may be used to strengthen weak flexor muscles of the rear limbs.

Exercise Considerations

Poles should be spaced at appropriate distances apart, determined by the dog's natural stride length. The distance between the rails should allow for a single step; the exercise may not be as beneficial if dogs are allowed to take two or three steps between rails. For muscle weakness, the rails are initially placed low to the ground. After the animal becomes accustomed to the task, the handler can further challenge the dog by adding more poles and increasing the height of all the poles to encourage greater active flexion and ROM of joints. Begin with walking 3-5 times over the rails, and progress to 8-10 times over the rails. Dogs may also progress to trotting for an additional rate of joint motion challenge. Leg weights may be added for additional strength training. Finally, for an extreme challenge, a gymnastic cavaletti ("bounces") can be created so that the dog must jump over each rail, land, briefly pause, and jump again without taking a stride between rails, creating a plyometric exercise.

Treadmill Walking and Jogging

Treadmills are excellent tools for helping to improve endurance, and they may be used to help with limb strengthening during gait. Not only can the walking or jogging time, length and speed increase, but the treadmill may be angled up or down to reduce or increase the forces placed on the forelimbs or rear limbs. Dogs that are trained to walk on leash generally are easily trained to walk on a treadmill. The flat, regular surface of the treadmill belt also allows exercise on a consistent surface with lower risk of a misstep. Patients should always be directly supervised, however, especially when the patient is becoming fatigued or the speed is at a fast walk or greater. In addition to monitoring for obvious fatigue, such as pulling on the harness or stumbling, it is possible to obtain respiratory rates and to record heart rates with an electrocardiogram machine or a telemetry unit to gain some insight regarding conditioning fitness over time. Care should be taken to not exercise beyond mild fatigue because the risk of injury may increase substantially.



Figure 30-18 To add resistance to limbs during treadmill activity to aid with muscle strengthening, an elastic band may be used to provide resistance to specific muscle groups.

Exercise Considerations

A warm-up period of walking and then fast walking is recommended prior to trotting on a treadmill. This also allows a period of adaptation to gaiting on the treadmill. It is important that the treadmill speed be rapidly increased when changing gait from a walk to a trot to avoid an awkward transition. In general, most medium to large dogs walk comfortably at a rate of 0.9-1.2 m/s (2-2.7 mph). A medium trot is generally 1.7-2 m/s (3.8-4.5 mph). To add resistance to limbs during treadmill activity to aid with muscle strengthening, an elastic band may be used to provide resistance to specific muscle groups (Figure 30-18). The band is secured to the involved extremity and the handler provides tension on the other end. For example, the band may be tied around the forearm with tension applied from the rear, providing resistance and strengthening of the elbow flexors and shoulder extensors during the swing phase of gait. Flexor muscle strengthening may be achieved by putting leg weights on the desired limbs.

Jogging

Jogging not only helps to improve cardiovascular conditioning, but it also helps with muscle strengthening. Jogging should be considered when increased speed is desired during the intermediate stage of recovery from injury or surgery, as a transition from walking to running. Early speed training aids muscle strength and the velocity of muscle contraction, important factors in improving speed. Consideration should also be given to the increased force placed on an affected limb when transitioning from walking to jogging. For example, walking places approximately 60% and 30% of body weight on a forelimb and hindlimb, respectively, at a walk.¹¹ Progressing to a medium-paced trot increases weight bearing to approximately 110% and 70% of body weight on a forelimb and hindlimb, respectively.¹²

The patient should not be exercised beyond mild fatigue. Exercising when fatigued leads to muscle weakness, poor control of accessory joint motion, and delayed proprioception responses, all resulting in possible joint injury.

Exercise Considerations

Jogging may be initiated in cases with stable surgical repairs when the dog is walking on the limb with minimal lameness and pain, and the stage of tissue recovery can withstand jogging. Begin slowly, jogging 0.5 to 3 minutes one to three times daily, and work up to 20 minutes two to three times daily. Be certain that lameness is not worse after jogging. As the patient develops adequate conditioning and strength to easily tolerate this level of activity, interval training may begin, which alters short bursts of speed activity with slower jogging. It is believed that this form of training may result in accelerated gains in speed and conditioning. An alternative if the therapist or owner cannot jog with the patient is to use an extendable leash to allow the dog to trot or jog out, and then return to the handler immediately, repeating this cycle several times for a total duration of 5-20 minutes.

Walking and Jogging in Tall Grass, Sand, or Snow

Walking or jogging a dog through a field of tall grass, sand, or snow enhances muscle strengthening, endurance, and cardiovascular conditioning because of the resistance provided by these media. In addition, dogs have a tendency to flex their joints to a greater extent as they negotiate these substances, resulting in additional cardiovascular conditioning and strengthening through a greater range of joint motion.

Exercise Considerations

Walking or jogging in deep sand or snow provides resistance to limb movement and strengthens the flexor muscles, which must contract more to lift the limbs. Extensor muscles may be preferentially recruited when walking or jogging in tall grass because they must contract to a greater degree to advance the limb. Exercising in sand and snow also minimizes concussive forces placed on arthritic joints, while allowing strengthening of supporting periarticular muscles. Further strengthening may be achieved by using leg weights, weighted backpack or elastic resistance bands.

Leg Weights and Resistance Bands

To add resistance to limbs during exercise and increase muscle strengthening, leg weights or elastic bands may be used to target specific muscle groups (Figure 30-19). Leg weights are frequently used in dogs with weakness as a result of injury to musculoskeletal tissues and subsequent surgery, or neurologic conditions. Leg weights may be fashioned from lead strips, or commercially available leg weights for people may be used. It is important that the



Figure 30-19 To add resistance to limbs during exercise and increase muscle strengthening, leg weights or elastic bands may be used to target specific muscle groups, such as the medial glenohumeral structures.

method of securing the weight is appropriate for dogs. Many leg weights designed for people may not fit properly. Many of these are filled with sand, and when the weight is applied to the thinner dog limb, the fastening straps may not provide a snug fit, allowing it to slip off. Removal of some of the sand may solve this problem. The leg weight must be firmly attached to the dog's limb because most dogs will attempt to shake the weight off when first introduced to it. In addition to considerations regarding securing the leg weight, placement on the limb should also be considered. The moment arm, and therefore the stress on the limb, is less when the weight is placed closer to the affected limb segment. If additional muscle force to move the limb is desired, the weight should be placed distal to the affected segment so that a larger muscle force is generated to move the limb and weight. A general rule of thumb is to use a weight that is between 1% and 3% of body weight, depending on limb strength and the stage of recovery. Therefore $\frac{1}{2}$ -lb leg weights may be used on dogs that weigh 10 to 20 lb, 1-lb weights on dogs weighing 20 to 40 lb, $1\frac{1}{2}$ -lb weights for 40- to 60-lb dogs, and 2-lb weights for dogs weighing more than 60 lb. Caution should be used when first applying the weights because some dogs may shake the limb or have exaggerated limb motion because of the altered sensation. It is possible that injury may occur, so it is important to gradually introduce the weight to allow a period of accommodation. Leg weights are most effective in strengthening the flexor muscles because these muscles are engaged to lift the limb during the swing phase of gait to advance the limb. In many dogs that are not accustomed to the leg weight, the limb will be held up in a flexed, non-weight-bearing position, and the therapist may be discouraged. However, the flexor muscles will fatigue in most cases, and the limb will be gradually

lowered to the ground. Strengthening of the flexor muscles has occurred in this situation, and eventually the dog places the limb on the ground, resulting in improved weight bearing while standing or walking. Therefore a positive outcome has occurred.

Elastic resistance bands are also frequently used for limb muscle strengthening. Bands come in various degrees of resistance, with the lighter colors generally having the least resistance and darker colors having greater resistance. The band is secured to the involved extremity and the handler provides stability on the opposite end. For example, the band may be tied around the forearm with the line of pull from the rear, providing resistance and strengthening of the elbow flexors and shoulder extensors during the swing phase of gait. Although leg weights require no real intervention on the part of the therapist other than securing the weight, the therapist is usually engaged while using resistance bands. The tendency is to try to "time" the movement of the limb with the therapist's hand motions. Frequently, this is mistimed and results in disruption of the gait cycle of the dog. In most cases, it is better to simply apply some light constant tension to the band and allow the dog to contract the desired muscle group against the band to move the limb. The tension is placed on the band in the direction opposite the muscle group for which strengthening is desired. For example, if strengthening of the quadriceps muscle is desired, caudal tension is placed on the band. Therefore when the limb is advanced and the quadriceps muscles contract to extend the stifle during the end of the swing phase of gait, the caudal tension on the band results in increased contraction strength of the quadriceps muscle. Unlike leg weights, which result in strengthening of all muscle groups, resistance bands may be used to target specific muscle groups. For example, in a dog with early medial shoulder instability, it may be desirable to strengthen the medial shoulder muscles, including the subscapularis. An elastic band may be used to increase the contraction strength of this muscle group. For example, while walking on a ground treadmill, gentle lateral tension is applied to the band, so that the adductor muscles must contract to a greater degree, resulting in strengthening of that group. Similarly, resistance bands may be used to apply tension while hopping the animal laterally, similar to hopping performed during neurologic evaluation. The degree of tension applied should not result in discomfort to the dog or significantly alter the dog's gait. In general, elastic resistance bands are used most frequently while walking over ground or on a ground treadmill, but they can also be used while swimming or walking on an underwater treadmill. In these situations, the band should be stretched out and laid flat to dry to avoid sticking together.

Pulling or Carrying Weight

A variety of harnesses are available for dogs to attach to carts or sleds for pulling weight. The harness should be



Figure 30-20 Pulling a sled with added weight helps with muscle strengthening.

well padded and comfortable. Pulling a cart with a large wheel diameter is easier than pulling a sled that slides along the ground. The positions of the head and neck are important in determining whether a dog pulls the weight forward with the forelimbs or drives forward from the rear limbs (Figure 30-20). If the dog carries its head and neck low to the ground, it is likely pulling with the forelimbs. A dog with the head and neck held high will shift some of the weight caudally and tend to use the rear limbs to drive the body forward. A variety of sleds and carts are commercially available. In all cases, the harness should be adjustable and properly fit the animal and cart or sled to avoid abnormal pressure where the harness contacts the animal; a poorly fitted rig may result in pressure sores and alter the manner in which the dog pulls the weight.

Dogs may also strengthen the core and postural muscles by wearing a canine backpack filled with weights. Weights may be loaded unequally or equally on both sides. When a dog wears a weighted backpack during down-to-stand or sit-to-stand exercises, greater muscle force is required to rise to a standing position. There does not seem to be a consistent pattern regarding placing weight equally on both sides of the dog or asymmetrically to enhance weight bearing on an affected limb. A trial-and-error approach should be used on an individual to determine the best placement of weight in the backpack to increase weight bearing on a particular limb. In general, these devices should be limited to dogs that are either healthy and using them for conditioning, or for dogs that are rehabilitating from a specific injury (e.g. CCL tear) and are using them for a specific purpose in rehabilitation, such as building up the hip extensor muscles. Dogs with conditions such as OA may be made worse by increasing the weight they are carrying, which increases forces on abnormal joint.

Aquatic Exercises

The reader is referred to Chapter 31 on aquatic therapy for specific information regarding aquatic exercise, especially underwater treadmill walking and trotting and swimming.



Figure 30-21 Additional strengthening of the limbs may be achieved by using flotation devices placed on a limb while walking on an underwater treadmill.



Figure 30-22 Resistance bands may be attached to a limb while walking on an underwater treadmill. The therapist pulls on the band in the direction opposite of the muscle group in which strengthening is desired.

Strengthening of muscle groups is achieved by moving the limbs against the resistance of the water. The effort to move the limbs increases by a factor of the velocity to the third power when the speed of swimming is increased; a similar phenomenon presumably occurs when the velocity of underwater treadmill walking increases. Further strengthening may be achieved by using flotation devices placed on the limbs (Figure 30-21). Similarly, resistance bands may be attached to the limbs; the therapist pulls on the band in the direction opposite of the muscle group in which strengthening is desired (Figure 30-22).

Speed Exercises

As the patient recovers and has very good to full weight bearing on the limbs, the goal should be to return the dog to as complete athletic function as possible. Although most dogs undergoing rehabilitation are not competitive

athletes, nearly all dogs enjoy running and playing. Therefore rehabilitation patients should be gradually conditioned to withstand the rigors of activity at moderate to high speeds. It is vital that this phase of training not occur before adequate tissue healing and muscle strength and endurance have occurred to prevent possible injury. Trotting, running, and running on a treadmill are all methods to increase speed. In addition, sport-specific activity should be gradually introduced to improve function. Interval training techniques may be especially effective in enhancing performance, including endurance training. With this form of training, near-maximal effort is used, followed by a less intense effort. The cycles are repeated during a single training session. The number and length of cycles are gradually increased to add conditioning stress to enhance performance.

Controlled Ball Playing

Ball playing is a fun and effective form of therapeutic exercise that dogs and their owners enjoy. It also has the potential to cause damage to surgical repairs. Controlled activity is the key. The degree of activity depends on the surgical procedure performed, the condition of the tissues, and the stage of tissue healing. Ball playing should begin on a relatively short leash to avoid explosive activity in the early postoperative period. As the patient progresses, the dog graduates to ball playing in an enclosed area, such as a run. As the animal nears full return to function, off-leash activity may be performed in a large fenced field free of irregular surfaces. The main benefits of ball playing are to increase power, speed, and muscle strength. In most conditions, jumping should be avoided to reduce the risk of injury. It is important to start by rolling the ball or throwing such that the ball is not bouncing high in the air, which encourages jumping.

Exercises for Cats

Physical rehabilitation of cats can be challenging. The principles of physiotherapy in cats are similar to those of dogs. However, the personality of some cats can alter the approach to physiotherapy. The most common reasons to perform physiotherapy in cats are generally related to injuries sustained as a result of trauma or joint conditions. Fractures of the long bones, especially the femur and tibia, as well as the pelvis are relatively common. Common joint conditions that undergo postoperative rehabilitation include cranial cruciate ligament rupture, deranged stifle, femoral head and neck excision, and patella luxation. On occasion, neurologic conditions occur in cats that could benefit from physiotherapy, including trauma and intervertebral disk herniation. Osteoarthritis is yet another condition for which physiotherapy may be beneficial.

Although most dogs improve and become more comfortable with rehabilitation as the sessions continue, most

cats may actually become less comfortable and more suspicious of various rehabilitation activities. In addition, many cats are not motivated by food rewards as dogs are. Therefore a different approach to physiotherapy is often necessary.

Proprioceptive exercises may be performed similar to dogs, and include weight shifting activities, balance board, wobble board, and standing on a therapy ball or roll. A small foam roll or roll of paper towels underneath the cat to help provide support may make the proprioceptive activities easier. The therapist must be very observant and watch the cat for signs of agitation to prevent the cat from having an unpleasant experience during rehabilitation. Initial sessions may only last for 30 seconds on a balance board, for example. If the cat has a pleasant experience, the sessions may gradually extend to 1 or 2 minutes on the balance board.

Most cats are not trained to walk on leash, but if a particular individual is able to, especially with the harness, then leash walks are some of the earliest therapeutic exercises that may be performed. Ground treadmill walking with support and assistance is another way to improve limb use. With many therapeutic exercises, cats that are accustomed to going into their carrier may be lured into an open door. This technique can be effective for a number of exercises. As the cat improves with treadmill walking, strengthening can be achieved with small leg weights or elastic resistance bands to increase the strength of muscle contraction. Many cats will resist this type of activity and it is important to be certain that the cats do not injure themselves while resisting.

Many cats will also play with feathers or toys tied on the end of the string to encourage activity and motion. A catnip mouse works well as a lure to get the cat to move. Cats that are conditioned to come running when the can opener is running can be exercised by putting them in a different area of the house and then starting the can opener, resulting in the cat coming to the open can.

Active ROM of joints is also important after many surgical procedures. One such exercise is to have the cat walk over low cavaletti rails. Again, an open carrier at the end of the cavaletti rails may entice the cat to walk over the elevated rails. It is important to let the cat win every so often by actually letting them advance into the carrier. Some cats respond quite well to a flashlight beam or laser light moving along the ground. Although cats tend to move in a relatively rapid manner as opposed to slow consistent movements, it can be an effective way to entice movements as well as stretching along a wall or vertical surface. However, caution should be exercised so that cats do not develop an obsessive-compulsive disorder with light chasing.

Aquatic therapy is possible in many cats. Although many cats do not like water, a number of cats will tolerate either swimming or walking on an underwater treadmill.

The therapist generally will need to accompany the cat in the aquatic environment. A small life jacket is advised in cats that will undergo swimming.

Although physiotherapy in cats is more challenging than in dogs, it is still an important part of recovery to help return them to full function as quickly as possible. Careful attention to behavior differences in cats and choosing activities that the cat enjoys are critical to successfully provide rehabilitation to cats.

Factors Affecting Progression of Exercises

Many variables must be considered in the design of a therapeutic exercise program to enhance strength, endurance, and speed, including the type and severity of the condition; the stability of any surgical repairs; the number of involved limbs and joints; the size of the animal; the preexisting physical condition of the patient; the available facilities; the skill and experience of the therapist; and whether the exercises will be administered by a professional, the owner, or a combination of the owner and a professional. Unfortunately, it is impossible to have a single protocol for each specific condition. Some basic guidelines may be applied, however.

The goal of any therapeutic exercise program should be to restore the animal to as full and active use as possible. The goal may need to be periodically adjusted if the patient progresses to a greater or lesser degree than expected. As the animal nears return to function, strategies should be developed to encourage muscle mass and strength, endurance or power, depending on the intended function of the patient.

It is very important to perform the exercises correctly and to maintain a consistent level of activity daily or every other day. When considering a muscle strengthening program, strengthening activities should only be performed two to three times per week to avoid overtraining and muscle pain. Overdoing activity will likely be detrimental to the animal's recovery. Appropriate nutrition is also important when undergoing muscle strengthening, especially appropriate protein and amino acid balance in the diet.

If pain or lameness increases during or several hours after stepping up an activity, the level of activity should be reduced by 50% for 3 to 7 days. If the lameness and pain then resolve, the activity may be slowly increased back to the pre-step-up activity level over 3 to 5 days, with careful observation to be certain that the activity is not increased too aggressively or rapidly. The rate of subsequent step-up activity should be half of what it originally was to avoid a recurrent episode of exacerbated lameness and pain.

It is also important to vary the routine for the animal's and the owner's sake to avoid boredom and to allow incorporation of function-specific activities in the animal's home or work environment. During chronic management

of the patient, weight control is extremely important. It is also important to continue to oversee the protocol and document improvement. Progress will occur at a slower rate during the chronic phase of recovery, but continued rehabilitation is just as important to achieve as full a return to activity as possible.

Several medications may be appropriate for use in the chronic rehabilitation period, including intermittent non-steroidal antiinflammatory agents and chondroprotective agents. Caution should be used, however, to be certain that any analgesic medications do not mask pain and discomfort associated with the level of activity. They should facilitate rehabilitation and recovery rather than allow rehabilitation to progress faster than normal conditions and tissue healing should allow.

Summary

Therapeutic exercises to increase joint motion, strength endurance, and speed are undoubtedly the most important aspects of rehabilitation. They may (and should) be incorporated with other modalities to enhance recovery, but it is the exercises that help improve muscle strength, joint mobility, limb use, and endurance to maximize function. Although a variety of techniques have been described, the ingenuity of the rehabilitation team, including the owner, will allow the development of other exercises that are specific to a patient's recovery. The keys to a successful therapeutic exercise program are to have site- and condition-specific exercises whenever possible, to use a variety of exercises and techniques to keep the therapy team and patient from becoming bored, and to allow the animal to appropriately progress so that tissues are adequately challenged for strengthening, but not so rapidly as to result in complications and tissue damage.

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31

Aquatic Therapy

David Levine, Darryl L. Millis, Jeffrey Flocker, and Lauren MacGuire

Aquatic therapy has become a very valuable component of veterinary physical rehabilitation. There are specific principles of aquatic therapy that provide unique benefits to the patient that other forms of therapy do not provide. In addition aquatic therapy comes naturally to many patients, which is especially beneficial because exercises in animal patients must be in a form that they naturally perform or can be easily trained to do. Although dogs cannot follow verbal instructions, they generally perform aquatic therapy very well.

Principles of Aquatic Therapy

It is necessary to understand the basic principles and properties of water, including relative density, buoyancy, viscosity, resistance, hydrostatic pressure, and surface tension to appreciate the benefits of aquatic therapy. These are important components to consider when planning an aquatic rehabilitation program.

Relative Density

The relative density of an object is the ratio of the weight of the object to the weight of an equal volume of water.¹ Relative density depends on the composition of an object. Densities of various substances are defined by a pure number value called *specific gravity*. The specific gravity of pure water is 1.0.² The specific gravities of fat, lean muscle, and bone are 0.8, 1, and 1.5 to 2, respectively.³ The relative density and specific gravity of an object determine how well an object will float. If the ratio of an object's specific gravity to that of water is greater than 1, the object will tend to sink, and if the ratio is less than 1, the object will tend to float.⁴ The specific gravity of an object also determines how much of the object's volume will float under water.⁴ For example, the specific gravity of ice is approximately 0.92; therefore 92% of ice will be submerged to displace enough water so that the upward force of buoyancy will equal the downward force of gravity, and 8% will float above the surface of water.^{2,5}

A lean person's specific gravity may be as high as 1.10, whereas an obese person's specific gravity may be as low as 0.93.⁴ An object whose specific gravity is equal to that of pure water (1) will float just below the surface of the

water. If a person's specific gravity is greater than 1, that person will tend to sink. The greater the specific gravity, the faster the sinking velocity.² Thus a lean animal that is not moving will have a tendency to sink faster than an obese animal.

Buoyancy

A body immersed in water is subjected to two forces, gravity and buoyancy. *Buoyancy* is defined as the upward thrust of water acting on a body that creates an apparent decrease in the weight of a body while immersed.² Buoyancy stems from Archimedes' principle: When a body is fully or partially submerged in a fluid at rest, it experiences an upward thrust equal to the weight of the fluid displaced.^{1,2,4,6} The amount of water displaced depends on the density of the body immersed relative to the density of water.² An immersed limb or body with a relative density less than that of water will be assisted to the surface by buoyant forces.⁷ An object with a relative density less than 1 floats because the weight of the object is less than the weight of the water displaced.^{5,8} Buoyancy acts directly through the center of buoyancy.³ The center of buoyancy is the centroid of the displaced volume of water and depends on the distribution of the displaced volume of fluid relative to the body.⁹ Problems arise when the center of gravity and the center of buoyancy are not in the same vertical line. This can occur with the use of improperly placed floating devices, which can disrupt the vertical line, causing an animal to tilt or flip over so as to reach a state of equilibrium.³ The location of the center of buoyancy varies depending on the percent of body fat and distribution of adipose tissue. The more caudal the distribution of adipose tissue (which tends to occur more often in human females than males), the closer the center of buoyancy is located to the center of mass.

Buoyancy aids in the rehabilitation of weak muscles and painful joints. It allows the patient to exercise in an upright position and may decrease pain by minimizing the amount of weight bearing on joints. For example the percentage of weight borne by human females when immersed in water is approximately 5.9% to 8.7% at the seventh cervical vertebral level, 25% to 31% at the level of the xiphoid, and 40% to 51% at the level of the anterior superior iliac

spine.¹⁰ Males bear 6.8% to 10%, 30% to 37%, and 50% to 56% of their body weight in the upright position at these same levels.¹⁰ Other studies of humans have also found similar percentages of weight bearing at these same levels.^{11,12} In a study performed on dogs the amount of body weight borne when immersed in water (as a percentage of body weight on dry ground) was approximately 91% when the water was at the level of the lateral malleolus of the tibia, 85% at the level of the lateral condyle of the femur, and 38% at the level of the greater trochanter of the femur (Figure 31-1).¹³ This information may be particularly useful when treating patients with arthritis because joints may be unloaded as a result of the buoyant properties of water.

Hydrostatic Pressure

Pascal's law states that fluid pressure is exerted equally on all surfaces of an immersed body at rest at a given depth. This fluid pressure is directly proportional to both the depth and density of the fluid.⁴ Therefore the deeper a body is immersed in water, the greater the pressure exerted. The atmospheric pressure at the surface of water is 1.00 kPa (15.7 psi), and fluid pressure increases by 0.029 kPa (0.43 psi) per foot of depth.³ It can be estimated that the pressure exerted on a part of the body immersed 3 feet below the surface will experience a fluid pressure of 1.09 kPa (17 psi). Because hydrostatic pressure provides constant pressure to a body or limb immersed in water, it may provide an improved environment for working with swollen joints or edematous tissues.⁷ Hydrostatic pressure opposes the tendency of blood and edema to pool in the lower portions of the body and can therefore aid in reducing swelling.³ The benefits of edema reduction because of hydrostatic pressure are used by pregnant women who exercise in water to help minimize swelling in their lower extremities during exercise. The reduction in edema is probably caused by several factors, including a general increase in circulation with exercise, and the prevention of further pooling of fluids in the lower extremities by the pressure exerted on them by water.

Hydrostatic pressure may also decrease pain during exercise. It is theorized that hydrostatic pressure provides phasic stimuli to the skin afferents (sensory receptors) that cause a decrease in nociceptor hypersensitivity. This acts to decrease a person's or animal's pain perception, which may allow the patient to perform a variety of movements with less pain.^{7,14}

Viscosity and Resistances

Viscosity is a measure of the frictional resistance caused by cohesive or attractive forces between the molecules of a liquid.¹⁵ The viscosity, or resistance to fluid flow, is significantly greater in water than in air,⁷ making it harder to move through water than to move through air. Water can therefore provide resistance that may strengthen canine

muscles and improve cardiovascular fitness. Viscosity is important in aquatic therapy for several other reasons. Viscosity may increase sensory awareness and assist in stabilizing unstable joints.⁷ It can also help prevent falling by increasing the time span for patients to react,⁷ which may reduce patient anxiety. For example dogs with paraparesis, obesity, osteoarthritis, or severe deconditioning may be more willing to walk in water than on land because of water's combined properties of buoyancy and viscosity, which help to support the dog.

Any body movement occurring in water must overcome water viscosity.¹⁶ The resistance created by the viscosity of the liquid is proportional to the velocity of movement through a liquid. With increased velocity of movement in water, there is a uniform increase in resistance.

Various movements in water may increase or help reduce resistance. Streamline flow is the steady, continuous movement of fluid. There is little friction between layers of fluid in streamline flow because the layers separate to move around the object and smoothly rejoin behind it.⁵ Turbulent flow consists of irregular movements of the layers of fluid.^{1,5} The irregular movement in turbulent flow causes increased friction between the molecules of the fluid and between the object and the fluid.⁵ Eddies are a form of turbulent flow in which the layers of fluid following a moving object in water move in circular directions.⁵ Eddies resist an object's movement in water by pulling the object backward. Because of all these factors, resistance in aquatic exercise may be increased by increasing the velocity of movement of the patient, increasing the surface area of the object or body part moving in water, or by increasing the length of the lever arm of the object moving in water.¹⁶

Surface Tension

The tendency for water molecules to adhere to each other is known as cohesion.² Water molecules have a greater tendency to adhere together on the surface. Resistance to movement is slightly greater on the surface of water because there is more cohesion on the surface of water.² Surface tension is the force of attraction between surface molecules of a fluid.³ Surface tension is not a factor if the moving body part is completely submerged in water. It becomes a significant factor when a limb breaks the surface of the water. Therapeutically if a patient is extremely weak, movements may be performed more easily in the water just beneath the surface rather than at or on the surface.

Aquatic Exercise Research

There are many benefits of aquatic therapy. Exercising in water is effective for improving strength, muscular endurance, cardiorespiratory endurance, range of motion (ROM), agility, and psychological well-being, while minimizing pain.¹⁷⁻¹⁹ The purpose of this section is to provide the reader

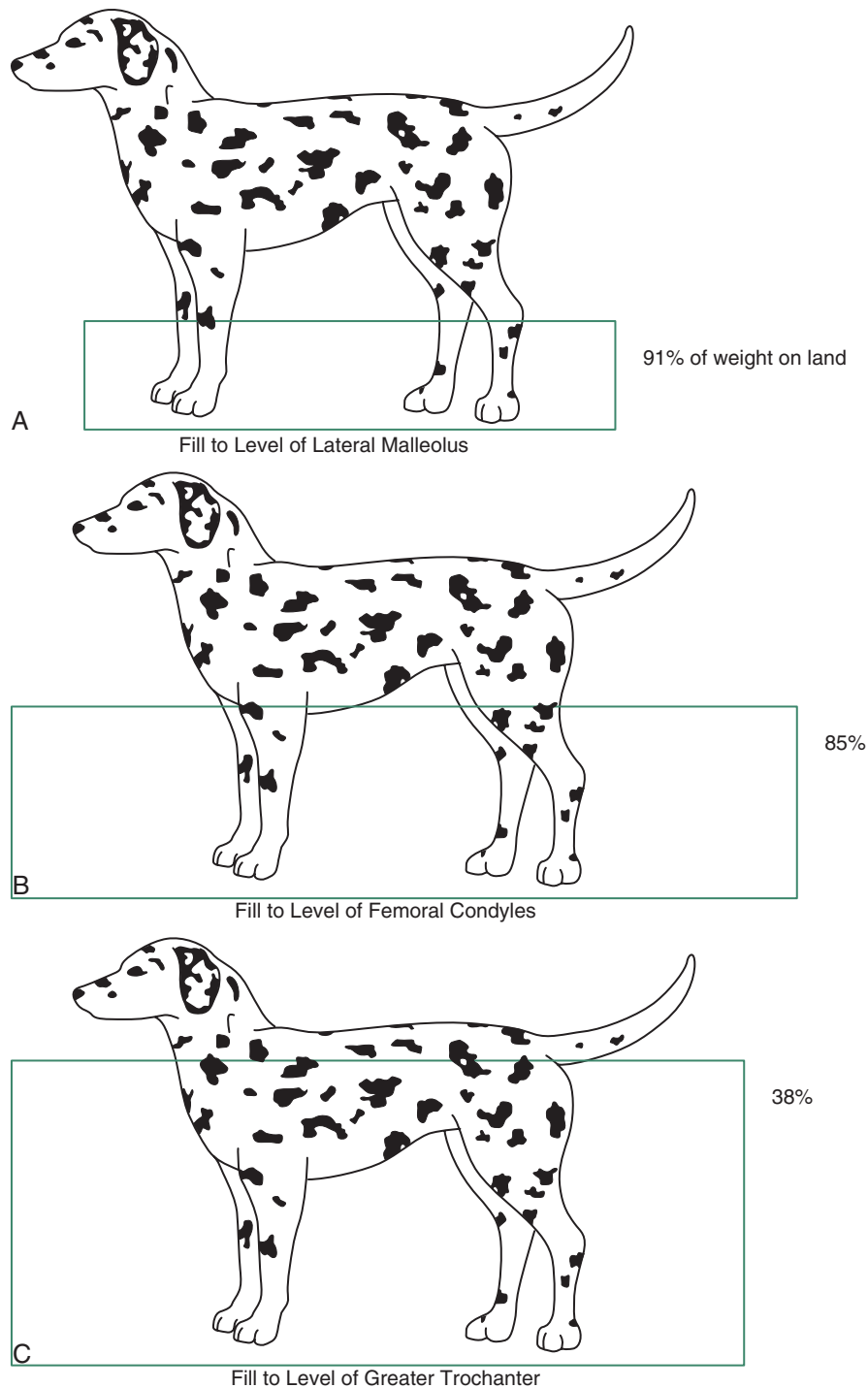


Figure 31-1 A, Dog in water to the level of the lateral malleolus. B, Dog in water to the level of the lateral epicondyle. C, Dog in water to the level of the greater trochanter. (Data from Levine D, Marcellin DJ, Millis DL et al: Effects of partial immersion in water on vertical ground reaction forces and weight distribution in dogs, *Am J Vet Res* 71:1413-1416, 2010.)

with information regarding human studies of the benefits of aquatic therapy. Table 31-1 provides a summary of these studies. It is likely that similar principles apply to animals.

Gravity is the primary resistant force to exercise on land, whereas viscosity, friction, and turbulence are the primary resistant forces to exercise in water; these properties have a direct effect on heart rate and oxygen uptake.²⁰ Arm exercises on land produced an increase in heart rate from a resting rate of 77 beats per minute to 92 beats per minute in men (age 21 to 27) and from 74 to 96 beats per minute in women (age 20 to 29). Exactly the same arm exercises in water produced an increase in heart rate from 60 to 99 beats per minute in men and from 60 to 98 beats per minute in women. Compared with the subjects exercising on land, the subjects exercising in water exercised at slightly higher heart rates and had a greater overall increase

in heart rate because of the initial lower resting heart rate. Subjects in this study had lower resting heart rates in water than on land because of two factors: the reflex response of the cardiovascular system to the cold receptors of the skin, and the effect of hydrostatic water pressure exerted on the submerged body.²⁰ The water temperature ranged from 26° to 26.5° C (78.8° to 79.7° F) during aquatic exercise. As discussed later, water temperature directly affects heart rate. Leg exercises further increased heart rate during water exercises: 76 (resting) to 124 (exercise) beats per minute for men on land versus 60 (resting) to 139 (exercise) beats per minute in water, and 73 (resting) to 126 (exercise) beats per minute for women on land versus 60 (resting) to 126 (exercise) beats per minute in water. The increased weight of legs as compared with arms may increase the necessary energy expenditure for exercises performed on land

Table 31-1 Summary of Human Studies on the Benefits of Aquatic Therapy

Study	Study Subjects	Study Parameters	Outcome
CARDIOVASCULAR STUDIES IN AQUATIC THERAPY			
Avellini et al., 1983	Three groups of five men	Subjects trained on cycle ergometer 5 days/wk for 1 hr at 75% $\text{VO}_{2\text{max}}$; group 1 exercised on land and groups 2 and 3 exercised in water at temperatures of 32° C and 20° C, respectively	Water groups exercised at a lower heart rate than land group
Bravo et al., 1997	77 postmenopausal women	Exercised in waist-deep water for 60 min 3 days/wk $\times$ 12 months	Improved functional fitness and psychological well-being
Choukroun and Varene, 1990	11 healthy volunteers (four women, seven men)	Subjects were immersed to neck in 25°, 34°, and 40° C water; cardiovascular demands measured at rest	Cardiac output increased significantly at 40° C versus lower temperatures
Evans et al., 1978	Six men	Walked and jogged at similar metabolic intensities on a treadmill in waist-deep water	Only one third to one half of the normal walking and jogging speed was required to work at the same level of energy expenditure in water as compared with on land (on a treadmill)
Hall et al., 1998	Eight women (mean 30.25 years old)	Submaximal exercise on land and on a treadmill in chest-deep water at water temperatures of 28° and 36° C	Heart rate higher at 36° compared to 28° C; walking in chest-deep water yields higher energy costs than walking at similar speeds on land
Johnson et al., 1977	Four men (mean 24 years old) and four women (mean 24.5 years old)	Same exercises performed by the same subjects on land and in water: shoulder ABD/ADD and hip flexion/extension with limbs extended	Decreased resting heart rate in water and increased VO_2 demands when exercising in water
Melton-Rogers et al., 1996	Eight adults with adult-onset rheumatoid arthritis (30-40 years old)	Graded maximal exercise test on a stationary bike on land and running in water wearing a flotation device	Higher maximum perceived exertion and respiratory rate were seen in water running; water provides a means for exercise for individuals with rheumatoid arthritis

Table 31-1 Summary of Human Studies on the Benefits of Aquatic Therapy—cont'd

Study	Study Subjects	Study Parameters	Outcome
Perk et al., 1996	20 stable COPD patients	Cardiorespiratory parameters were measured on land and in temperature-controlled water (32°C) at rest and during a 15-min submaximal upper-body exercise program	At rest in water there was a decrease in systolic and diastolic BP
Whitley and Schoene, 1987	12 healthy female college students	Heart rate was measured immediately after walking in waist-deep water and then on treadmill at same speed, duration, and distance	Heart rates were higher at each speed when walking in water vs. walking on land
RANGE OF MOTION AND STRENGTH STUDIES IN AQUATIC THERAPY			
Suomi and Lindauer, 1997	17 women with arthritis	Exercise subjects participated in three 45-min aquatic training sessions/wk for 6 wk; control subjects were asked to refrain from participating in any organized physical activity	Greater increase in strength and ROM occurred in the aquatic therapy subjects
Tovin et al., 1994	14 men and six women (16-44 years old) with ACL reconstructions	10 subjects participated in traditional rehabilitation protocol (stationary cycling, gait training, steps, hip flex/ext, ABD/ADD in standing, and knee flexion in sitting), whereas other 10 subjects performed same exercises in a pool; performed over first 8 postoperative wk	Rehabilitation program for patients with intraarticular ACL reconstruction performed in a pool was more effective in reducing joint effusion and facilitating recovery of lower extremity function as indicated by Lysholm scores; rehab in water is equally effective as on land for restoring knee ROM and quadriceps strength

ABD, Abduction; ACL, anterior cruciate ligament; ADD, adduction; BP, blood pressure; ROM, range of motion.

because of the increased mass and pull of gravity. This study demonstrated that heart rate and oxygen uptake were greater when performing exercises in water compared with performing the same exercises on land.²⁰

The metabolic requirements are also greater for exercises performed in water than for exercises performed on land. In a study by Johnson et al.,²⁰ men and women performed the same arm and leg exercises in water and on land. Oxygen uptake during exercise in water was 34% greater in men and 27% greater in women as compared with the same exercises on land.²⁰ The differences between men and women are probably owing to different body compositions (male body composition tends to be more muscular, which requires more oxygen) and heights (men tend to be taller and have longer extremities, thereby having a longer lever arm, which requires more exertion to move). Another study found that subjects could walk or jog at slower speeds in water than on land while expending the same amount of energy as measured by oxygen consumption. In this study subjects walked at 2.6 to 3.5 km/hour (1.6 to 2.2 mph) in water, whereas subjects on land walked or jogged at speeds of 5.5 to 13.4 km/hour (3.4 to 8.3 mph).

Both expended the same amount of energy, despite the greater velocity of people walking on land.²¹ A study by Whitley and Schoene¹⁹ supports these data with the observation that various types of exercises performed in water (aquatic calisthenics, shoulder and leg exercises, bicycle ergometry exercises, walking, and jogging) require a significantly greater amount of energy expenditure compared to the same exercises performed on land. Subjects in a study by Hall et al.²² also expended more energy walking in water than on land at speeds greater than 4 km/hour (2.5 mph).

When aerobic exercise is used in the rehabilitation process, it may aid in improving cardiovascular fitness and weight reduction. Melton-Rogers et al.²³ compared exercise during land bicycle ergometry with running in water with a flotation device in individuals with rheumatoid arthritis. Higher ratings of perceived exertion (RPE) by the subjects and respiratory exchange ratios were seen during water running.²³ These studies demonstrate that walking in water at specific speeds may be sufficient to meet heart rate requirements for developing and maintaining cardiorespiratory fitness.

Aquatic exercises are also beneficial for muscle strengthening. Bravo et al.¹⁷ performed a study on postmenopausal women who exercised in a pool with waist-deep water for 60 minutes, three times a week for 12 months. The average strength gain as a component of functional fitness was approximately 18%. Because of buoyancy water walking can provide an excellent alternative method of achieving and maintaining muscle strength and ROM in individuals with lower limb joint problems such as osteoarthritis (OA). Although exercise in water may not be as effective in achieving maximum muscle performance as land exercises, rehabilitation in water may minimize the amount of joint effusion and lead to greater functional improvement.²⁴ Closed-chain exercises such as walking in water are more closely associated to movements in everyday activities than open-chain exercises because they recruit muscles in a more functional manner.²⁵ Performing closed-chain exercises in an environment in which weight-bearing forces are decreased may minimize or eliminate damage and inflammation to the soft tissues, while maximizing functional training. The aqueous environment may also aid in reducing knee pain and joint effusion,²⁴ which may facilitate the recovery of lower-extremity function after cranial cruciate ligament (CCL) stabilization in dogs.

Water's buoyancy can ease the performance of exercise activities while also providing proprioceptive feedback to aid in the rehabilitation process. The effect of buoyancy allows for gentler active exercises by decreasing the loads placed on the injured tissues compared with exercises performed on land.¹⁸ Aquatic exercises may be used as a transition to land-based exercises in postsurgery or postinjury rehabilitation. Water exercises are generally less painful than land exercises because of the support that buoyancy provides. Therefore water exercises may result in less discomfort maintain ROM and functional movement, and provide a better sense of security when initiating active movements.¹⁸

There are many physiologic effects resulting from exercise in heated water. Among them are increased circulation to muscles, increased joint flexibility, and decreased joint pain.²⁶ Templeton et al. performed a study to determine whether subjects with rheumatic diseases experienced changes in joint flexibility and functional ability following an aquatic therapy program.²⁶ Subjects exercised in water for 8 weeks. Exercises consisted of therapeutic games emphasizing both isotonic and isometric exercises of the trunk, neck, and upper and lower extremities. Water temperature was maintained at 32.9°C (91.3°F), and the air temperature was maintained at 31.7°C (89°F). Active ROM and a Functional Status Index test that recorded the degree of assistance, pain, and difficulty experienced in the performance of 18 tasks were measured before and after the 8-week aquatic therapy session. The results of the study indicated that pain decreased with therapy, and active ROM and functional ability increased. The decreased pain and

improvement in performing daily activities had a significant impact on the overall increased functional ability of the subjects.²⁶ These findings support aquatic therapy as an effective rehabilitation technique for increasing functional ability and joint flexibility in people with severe joint disease. Osteoarthritic and rheumatoid arthritic patients participated in another study of the effects of aquatic exercise on ROM and strength.²⁷ Control subjects were asked to refrain from participating in any new physical activity throughout the duration of the study, whereas the exercise group participated in an aquatic exercise program. After 6 weeks, hip strength and ROM were measured. Hip strength in the aquatic exercise group increased 10.9%, whereas ROM increased 11.8% as compared with the control group.²⁷

Water temperature may have a significant effect on the cardiovascular response to exercise. Subjects in cool water exercise at lower heart rates than those exercising on land at the same workload for several reasons.²⁸ If the water temperature is low, peripheral vasoconstriction occurs, blood moves centrally, venous return is enhanced, and stroke volume increases. Therefore to maintain the same cardiac output, heart rate decreases. In one study, subjects training in cool water (20°C [68°F]) had enhanced stroke volume and decreased heart rate, thereby increasing exercise efficiency.²⁸ It is also our experience that dogs exercise at lower heart rates walking on underwater versus land treadmills at the same velocity and length of time (unpublished data, Millis DL, 2000).

Blood pressure is also affected by water temperature. In a study of subjects at rest while submerged in 32°C (89.6°F) water, systolic blood pressure decreased 14 ± 14 mm Hg and diastolic pressure decreased 6 ± 10 mm Hg because of peripheral arterial dilation. On the other hand, exercising in water at temperatures that exceed body temperature may increase cardiovascular demands above those of exercise alone.¹² In one study, cardiac output was measured at rest while people were immersed in water temperatures of 25°, 34°, and 40°C (77°, 93.2°, and 104°F).³⁰ Cardiac output of those immersed in 40°C water increased significantly. The authors of this study recommend that elite athletes train in water temperatures between 26°C and 28°C (78.8°F and 82.4°F) to prevent any heat-related complications.³⁰ This temperature range is also recommended by the authors for pools in which healthy dogs are training. Lower temperatures are also apparently tolerated quite well by dogs, especially those with thicker hair coats.

All of the benefits of aquatic therapy in humans (improved strength, muscle endurance, cardiorespiratory endurance, increased ROM, decreased stress on healing tissues, and minimized pain) may also apply to dogs. The type of aquatic therapy used by therapists, veterinarians, and owners depends on the specific rehabilitation needs for each individual. An understanding of the benefits of aquatic therapy aids the prescription of specific protocols.

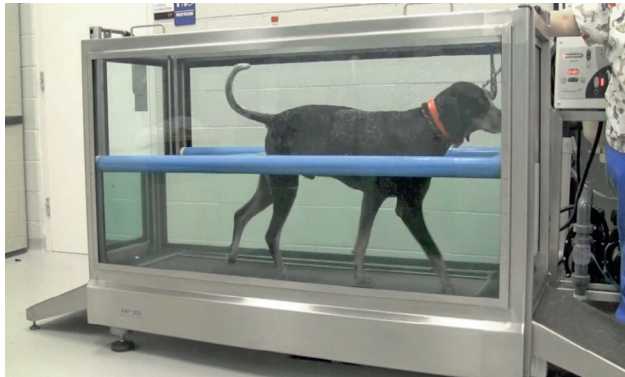


Figure 31-2 Dog walking on underwater treadmill.

Aquatic Therapy Equipment

This section provides the reader with information concerning the types of aquatic therapy and equipment currently available for dogs. The anticipated use, size of the dogs receiving therapy, and funds available help determine the type of equipment best suited for rehabilitation. Some of the available forms of aquatic therapy include underwater treadmills, pools of various sizes, and whirlpools (Figure 31-2).

Underwater treadmills are very useful for aquatic therapy. In addition to the buoyancy and resistance effects of water, the treadmill encourages walking or jogging with a more normal gait pattern than is achieved with swimming. The height of the water can be adjusted, which alters the buoyancy and alters joint motion. Underwater treadmills are very useful in neurologic patients to help with gait patterning with the support of the water. Units typically have entrance and exit doors, a storage tank for water, heater, filters, and water treatment with chlorine or bromine. Dogs typically walk into the unit, the doors are sealed shut, and the water is pumped into the chamber until the desired height is reached. The treadmill is then turned on at the appropriate speed. On occasion, a therapist may accompany a patient to assist with gait training. After treatment is completed, the water is pumped out, the doors are opened and the patient is rinsed off and dried. Units are available in some countries that have the water in the treadmill compartment, with an adjustable treadmill that lowers into the water. This type of unit helps to save space because a storage tank is not necessary, but the dog must be elevated on a lift to step on the treadmill; therefore, caution must be used to avoid the danger of the dog jumping from the lift or treadmill.

Swimming pools may also be used for rehabilitation. Several options must be considered when installing a pool. Some of the considerations include whether jets or propellers used to circulate the water and add resistance to exercises will be useful; the location of the jets; the possibility of adding an underwater treadmill into the pool; hooks on



Figure 31-3 Dog receiving aquatic therapy in whirlpool.



Figure 31-4 Dog wearing life vest.

the inner walls to which safety straps may be attached; manual versus computer-controlled water treatment and heating systems; above-ground versus in-ground pools; and stairs, lifts, or ladders to enter the pool. In addition the physical aspects of the enclosure and surrounding area must be considered. For example the humidity and condensation associated with a heated indoor pool must be considered. The floor surface around the pool must also be carefully designed to avoid slipping and falling by both patients and handlers.

Other forms of aquatic therapy include hot tubs or whirlpools used in hospitals (Figure 31-3). Cost-effective forms of aquatic therapy for smaller dogs include bathtubs, large basins, and plastic pools. Taking a dog to a lake or river is also a cost-effective means of providing aquatic therapy, but caution must be exercised when swimming dogs with recent incisions in these environments. Strong water currents and natural hazards must also be avoided.

Equipment used in aquatic therapy for dogs can be relatively simple and inexpensive. One useful piece of equipment is a pet life preserver (Figure 31-4). These are designed to support the dog and allow free breathing. They



Figure 31-5 Using a life vest to guide a dog during pool exercises.



Figure 31-6 Dog being transferred into a pool using a lift. (Courtesy Trish Penick, San Diego, CA.)

may include quick-release buckles with adjustable straps and handles for an attendant to assist the dog while swimming. Various sizes are available. They are also useful to control and guide the dog while performing pool exercises (Figure 31-5). Other devices used in aquatic therapy are balls and sticks for fetch games, a lift for entering or exiting the water (Figure 31-6), and harnesses to direct and maintain contact with the dogs if the attendant is not in the pool.

Persons working in the water with dogs may also benefit from specialized equipment such as wetsuits or other protective clothing to guard against prolonged exposure to pool chemicals or cold water (see Figure 31-6). Those working from outside the pool with dogs in water should also expect to get wet and should take precautions.

The equipment required to clean and maintain tanks and pools includes disinfectant solutions, protective gloves, buckets, a water hose, brushes with handles to enable scrubbing of crevices, and clean, dry towels.² According to Hecox et al., there is no conclusive evidence regarding clear advantages of any one specific type of disinfectant over another.² Studies and reported observations suggest sodium hypochlorite (5.25%), sold as household laundry bleach, is an inexpensive and effective disinfectant. However, bleach may corrode the surface of stainless steel

over time.³¹ Although manual scrubbing with standard germicidal detergents may reduce bacterial contamination, complete removal of bacteria from the drains, edges, and bottom of the tank may not be possible.²

Chlorine or bromine are commonly used to sanitize the water in pools and underwater treadmills. These levels are checked on a daily basis and adjusted as needed. Occasionally a reaction such as a skin rash occurs but by rinsing dogs off with water after aquatic therapy this is minimized. Other values such as the pH and alkalinity are also checked and adjusted. Dogs that defecate or urinate in the water or are incontinent pose additional problems in maintaining the cleanliness of tanks or pools. If an animal eliminates or is incontinent in the water, the best way to minimize cross-contamination is to drain the pool after each use and sanitize it. It is also recommended that this precaution be taken with dogs that have open wounds. If draining the pool is not possible because of its size, the pool should be shocked with a chemical disinfectant, and dogs that have open wounds should not use the pool for a period of time recommended by the manufacturer to ensure that the water has been sanitized. Although soiling the water poses potential problems, screening dogs before aquatic therapy may help to reduce or avoid this problem. Dogs should be walked before engaging in aquatic therapy to allow them an opportunity to eliminate.

Aquatic Therapy for Dogs

Conditions Benefiting from Aquatic Therapy

There are many conditions for which aquatic therapy may be beneficial, including rehabilitation of postoperative fractures, CCL stabilization, neurologic conditions, tendinitis, conditioning, and other disorders in which a dog is reluctant to use the limb or there is a lack of strength, ROM, proprioceptive ability, or weight-bearing status.³² The potential benefits such as strengthening, cardiovascular endurance, and improved function have been documented in humans, and aquatic therapy is now becoming more common in veterinary medicine. Outcome assessments of aquatic therapy programs are important for documenting progress and to determine the effectiveness of programs.

Contraindications and Precautions to Aquatic Therapy

Some dogs fear water or are reluctant to swim, and this must be considered before attempting treatment. In general, dogs tolerate an underwater treadmill better because the water rises from the bottom rather than having the dog enter or be lowered into the water, such as a swimming pool. If a dog panics the dog and handler could be placed in a potentially dangerous situation. The dog may injure itself with excessive thrashing. The handler may be injured

trying to restrain a panicked dog, or may also risk attack by the animal. Questioning the owner regarding a dog's history of swimming is critical, in addition to prudent safety and handling techniques. If possible, dogs that will be receiving aquatic therapy following nonemergency surgical procedures should be introduced to aquatic therapy before surgery. This allows for evaluation of the dog's disposition toward swimming and also allows the dog to be introduced to the therapy received postoperatively. Preoperative evaluation is not possible for some traumatic injuries, such as fractures. When receiving aquatic therapy, dogs should never be left unattended in the water.³³

Some veterinarians believe it is better to wait until incisions or open wounds are healed before placing the animal in water to minimize the risk of infection; others may begin aquatic therapy after the incision is sealed but before the incision is completely healed. A safe recommendation is to wait until after suture removal if there are no wound complications, including discharge from the incision, gapping of the wound edges, or any evidence of infection. However, there may be conditions that warrant earlier use of aquatic therapy. Regardless of when therapy is started, the wound should be sealed before initiation. The cleanliness of the water may influence this decision, as well as the animal's overall health and medical history. Betadine impregnated adhesive drapes may be also be used to help seal an incision area prior to entering the water to protect incisions.

The cardiovascular fitness of the dog must be considered because many dogs are initially unable to swim more than a few minutes before fatiguing. Swimming the dog several times daily for only 2 to 5 minutes may still result in significant gains in strength, ROM, function, and overall cardiovascular fitness.^{33,34}

Aquatic Therapy Basics

Reluctance to use a limb following injury is a common problem. Dogs that are reluctant to use a limb on land may use it in water to swim or walk on the bottom. Dogs that may benefit from extra buoyancy because of weakness may be assisted by placing a life jacket or other flotation device around the dog's trunk, or by the use of an overhead harness.³³ Owners may provide additional motivation to encourage dogs to walk or swim. Having the animal fetch a ball or other object may have the same motivational effect. If possible swimming with the dog is another excellent way of encouraging therapy, in addition to ensuring that someone is present if the animal has any trouble. Caution must be used when swimming injured dogs in lakes and rivers. Dogs must be prevented from diving or lunging off of docks or the shore when entering the water to prevent injury. Wading into the water with the dog until the dog is buoyant is recommended. Maintaining contact with the dog at all times is important. In addition, water sanitation and natural hazards must be considered.

Walking in the water, either on the bottom of the pool or on an underwater treadmill, is an excellent way to allow a dog with painful joints to exercise more comfortably because of the buoyancy provided by the water. This type of exercise may be used as a progression to swimming if the animal is either too weak or in too much pain to initially swim.

In addition to the buoyancy effects of swimming or walking in water, joint kinematics are also altered during aquatic therapy. Knowledge of aquatic kinematics is useful in designing specific exercise programs. For example in one study comparing swimming with ground treadmill walking, dogs that swam had significantly greater stifle flexion and total stifle ROM compared with ground treadmill walking (Table 31-2).³⁵ Marsolais et al. published data supporting swimming as an effective rehabilitation modality in dogs following extracapsular repair of cranial cruciate rupture. Two groups of dogs, control and surgical, were enrolled in the study. The surgical group consisted of 26 adult dogs that underwent debridement of a ruptured CCL, partial or complete medial meniscectomy, and extracapsular stabilization. The control group consisted of 26 healthy dogs without cruciate injury. All of the dogs in the surgical group underwent the same aquatic rehabilitation program focused on swimming. The rehabilitation sessions were performed twice daily, 5 days per week and consisted of 10 minutes of limb massage and passive ROM, 10 minutes of walking on a treadmill, and 10 to 20 minutes of pool time. Pool time consisted of alternating 1 minute of swimming and 1 minute of rest for a total of 5 to 10 minutes of swimming. None of the dogs were allowed to touch the floor of the pool during the swimming session except when entering and exiting the pool. The owners of the control dogs were instructed to allow normal activity several days before data collection. The control dogs did not participate in any part of the rehabilitation program. Data collection consisted of measuring hip, stifle, and hock flexion and extension with slow (0.9 m/s) and fast (1.3 m/s) walking on a treadmill and while swimming. Data collection revealed increased hip, stifle, and hock flexion ROM in the control group during swimming over walking on the treadmill and increased stifle (50° swimming and 30° walking), and hock (50° swimming and 35° walking) flexion ROM in the surgical group during swimming. Stifle and hock extension, however, was less when swimming compared with walking in both groups. Hip joint ROM was not affected in the surgical group when swimming compared with walking; however, the control group demonstrated increased hip ROM with swimming over walking. The authors summarized that swimming is an effective modality to gain stifle and hock flexion following extracapsular stabilization of CCL because it allows early mobilization without the added stress of weight bearing. Peak vertical forces were 18.5% higher in the rehabilitation group 6 months after surgery.³⁶

Table 31-2

Results of Kinematic Evaluation of the Hip, Stifle, and Tarsal (Hock) Joints During Swimming, Fast Walking (1.3 m/s), and Slow Walking (0.9 m/s) in Healthy Dogs (Control; n = 13) and Dogs (7) That Had Undergone Surgical Management of a Ruptured Cranial Cruciate Ligament and Were Enrolled in a Postoperative Aquatic Rehabilitation Program

Joint	Variable	SWIMMING		FAST WALKING		SLOW WALKING	
		Control	Rehabilitation	Control	Rehabilitation	Control	Rehabilitation
HIP	Maximal flexion (°)	83.6 ± 15.1 ^a	75.6 ± 7.9 ^d	95.9 ± 11 ^a	104.5 ± 12.5 ^e	98.6 ± 11.4 ^a	104.7 ± 13.8 ^e
	Maximal extension (°)	128.2 ± 14.4 ^a	114.9 ± 11.7 ^{d,*}	130.5 ± 8.3 ^a	138.4 ± 15.1 ^e	130.7 ± 10.7 ^a	137.1 ± 18.6 ^e
	Angular velocity (°/s)	160.9 ± 68.1 ^a	122.7 ± 47.0 ^d	88.2 ± 15.0 ^b	87.4 ± 21.8 ^e	62.8 ± 15.3 ^c	68.7 ± 11.3 ^f
STIFLE	Maximal flexion (°)	46 ± 10.7 ^a	46 ± 6.3 ^d	97.1 ± 8.7 ^b	100.1 ± 7.1 ^c	101.5 ± 8.0 ^b	102 ± 9.3 ^e
	Maximal extension (°)	112.9 ± 11.8 ^a	94.4 ± 16.8 ^{d,*}	133 ± 7.7 ^b	130 ± 9.3 ^c	135.1 ± 9.2 ^b	129.4 ± 10.6 ^e
	Angular velocity (°/s)	151.5 ± 39.3 ^a	94.1 ± 19 ^{d,*}	182.3 ± 50.9 ^a	138.9 ± 48 ^{d,*}	154 ± 38.2 ^a	105 ± 47.6 ^{d,*}
HOCK	Maximal flexion (°)	52.9 ± 12.1 ^a	74.7 ± 7.8 ^{d,*}	107 ± 7.3 ^b	120.2 ± 13.7 ^e	111.9 ± 7.8 ^b	124.8 ± 14.3 ^e
	Maximal extension (°)	127.7 ± 10.8 ^a	126.8 ± 12.6 ^d	141.6 ± 8.7 ^b	153.8 ± 12.4 ^{e,*}	141.9 ± 8.7 ^b	152.7 ± 11.3 ^{e,*}
	Angular velocity (°/s)	216.9 ± 62.7 ^a	134.1 ± 48.9 ^{d,*}	80.9 ± 20.7 ^b	70.5 ± 10.8 ^e	55.8 ± 14.8 ^c	48.6 ± 8 ^f

Data are given as mean ± SD.

^{a,b,c}For dogs in the control group, values in each row with different letter superscripts were significantly ($P < .05$) different.

^{d,e,f}For dogs in the rehabilitation group, values in each row with different letter superscripts were significantly ($P < .05$) different.

*Significantly ($P < .05$) different from corresponding value for control group.

From Marsolais GS, McLean S, Derrick T, et al. Kinematic analysis of the hind limb during swimming and walking in healthy dogs and dogs with surgically corrected cranial cruciate ligament rupture. *JAVMA* 222:739-743, 2003.

Table 31-3 Conditions for Achieving Maximum and Minimum Joint Flexion and Extension during Walking on Either a Ground or an Underwater Treadmill

Joint	MAXIMUM JOINT MOTION		MINIMUM JOINT MOTION	
	Extension	Flexion	Extension	Flexion
FORELIMB				
Shoulder	142 degrees—ground	93 degrees—water	132 degrees—water (level 1)	125 degrees—ground (level 1)
Elbow	159 degrees—water (level 3)	70 degrees—water (level 3)	124 degrees—ground	98 degrees—water (level 4)
Carpus	193 degrees—water (level 4)	97 degrees—ground	185 degrees—water (level 1)	119 degrees—water (level 4)
HINDLIMB				
Hip	141 degrees—ground	93 degrees—water (level 2)	120 degrees—water (level 1)	115 degrees—ground
Stifle	141 degrees—ground	64 degrees—water (level 2)	113 degrees—water (level 1)	109 degrees—ground
Hock	156 degrees—water (level 3)	81 degrees—water (level 2)	135 degrees—water (level 1)	115 degrees—ground

Values indicate the mean maximum and minimum joint flexion or extension angles of 8 dogs that occur under the indicated conditions during walking.

Ground, Walking on a ground treadmill; *water*, walking on an aquatic treadmill; *level 1*, water level at the greater trochanter; *level 2*, water level at the stifle joint; *level 3*, water level at the hock joint; *level 4*, water level at the bottom of the foot.

From: Jackson A, et al.: *Joint kinematics of dogs walking on ground and aquatic treadmills*. In Proceedings of the 2nd International Symposium on Rehabilitation and Physical Therapy in Veterinary Medicine, Knoxville, 2002.

However, stifle extension was less in swimming dogs than in dogs undergoing ground treadmill activity. In addition dogs had less stifle flexion, extension, and total ROM following stifle surgery as compared with normal dogs.

Rehabilitation of dogs on an underwater treadmill may provide additional benefits regarding joint kinetics. The relative degree of buoyancy and resistance to walking in water can be adjusted by changing the water level. In addition, adjusting the water level to the level of various joints may further influence joint kinematics because those particular joints must gain the necessary momentum and force to break through the water surface and overcome surface tension of the water. One study compared dogs walking on a ground treadmill to dogs walking on an underwater treadmill at the same velocity (Table 31-3).³⁷ In addition the water level was adjusted to the greater trochanter (level 1), stifle (level 2), hock (level 3), or the bottom of the foot (level 4). Joint motion was determined in both front and hind limbs, with each limb in a fully flexed non-weight-bearing position, with each limb in the final propulsion stage of weight bearing, and with each limb in the initial braking stage of weight bearing. Unlike swimming there was no significant reduction in joint extension with underwater treadmill walking during the initial braking phase of stance with water levels at and distal to the stifle. Therefore dogs had full active extension of all joints during a limb cycle, unlike joint motion during swimming. However, hip, stifle, and hock extension during the late phase of

propulsion was decreased when water was filled to the level of the greater trochanter. Joint flexion was increased with underwater treadmill walking. In general joint flexion was greatest with water levels at or higher than the joint of interest. Therefore the advantages of additional joint flexion achieved with swimming are also present with underwater treadmill walking, but full extension of joints also occurs with treadmill walking, which is not the case with swimming.

A flotation device, such as a children's water wing, applied to a limb may also be used to alter joint kinematics. For example, if additional motion of all joints of a particular limb is desired, the device may be placed on the distal limb. If additional joint extension and increased stance time are desired, the device may be placed on the contralateral limb. The altered sensation and unbalanced buoyancy of opposite limbs contribute to altered joint kinematics and may be used to achieve certain goals during rehabilitation.

Underwater Treadmill Therapy

Aquatic therapy performed in an underwater treadmill allows for active range of motion similar to that in a normal gait using the warmth, buoyancy, and resistance of water. Underwater treadmill therapy is used when there is a need for walking or jogging using a relatively normal gait pattern, with increased muscle or cardiovascular demand and minimal risk of injury. Secondly, it may be used to

reduce muscle spasm or peripheral edema and improve mood or bolster confidence.¹⁻³

Advantages

Activity in an underwater treadmill has several advantages over swimming or immersion in water. Dogs move in a natural walking or jogging gait pattern, and joint extension is more complete than with swimming. The speed of the treadmill may be adjusted. Slow speeds are used for early postoperative conditions, moderate neurological conditions, and patients beginning a conditioning program. Faster speeds may be used for higher levels of conditioning. The level of the water may be adjusted to alter the degree of buoyancy. This may reduce the risk of injury and reduces joint concussive forces. The resistance to movement through water increases the activity of muscles during the swing phase of gait. Warm water temperatures increase circulation, caloric expenditure, and nerve conduction velocity; improve coordination; and allow for increased soft-tissue elasticity.¹⁻³ Additionally, hydrostatic pressure exerted by the water on extremities aids in the reduction of edema.⁴

Underwater treadmill therapy allows for earlier intervention as compared with swimming. It promotes earlier active range of motion of joints especially in flexion, and strengthening of muscles. Functional carryover to land-based activities is usually seen very quickly.

Underwater Treadmill Temperature

Most underwater treadmills are maintained between 78 to 90 F. Warmer temperatures may be beneficial for relaxation of muscles. Cooler temperatures allow for more vigorous exercise with less risk of overheating. When treating a neurologic patient warmer water (85° to 90°F) may be preferable. Orthopedic patients and dogs in conditioning programs do not require the same warm temperatures (78° to 83°F).

When to Initiate Underwater Treadmill Therapy

Determining when therapy should begin is a decision made by the surgeon or veterinarian overseeing patient care. Following most orthopedic surgeries, 5 to 7 days is adequate if the incision is sealed with no drainage and the wound edges are well apposed. The incision may be covered with an adherent water-resistant bandage. However, some elect to wait until the time of suture removal to avoid possible incisional complications. This allows for sufficient initial tissue healing and initial therapeutic exercises. For surgeries such as total hip replacements, 4 weeks or more may be recommended. Neurologic rehabilitation postsurgery varies depending on the procedure and the quality of the stabilization. Particular care should be taken to support the patient and not allow the incision to become wet. Dogs with conservatively managed intervertebral disk disease

should not have therapy prior to 1 month after injury. These are common time frames for early aquatic intervention and vary from patient to patient.

Equipment

Before getting in the water with the patient, precautions should be taken on the part of the therapist. Patient may be fitted with a flotation vest to provide additional buoyancy. Be certain that the personal flotation device (PFD) conforms well to the body and does not interfere with either shoulder extension or hip flexion. Therapists may want to wear either a shortie or full-length wetsuit depending on their skill level and ambient temperature above the water. Wearing a wetsuit minimizes injury by flailing paws or dogs climbing onto the therapist. It is also important to have a supply of float toys, balls, and assistive devices like cuffs made from pool noodles, weight cuffs, elastic bands, rubber bands, and/or hair scrunchies easily accessible. These are used to encourage different range of motion of the affected or unaffected limbs. Pool noodles may also be placed along side, in front or behind the animal to keep it centered while walking on the treadmill.

Transfers

Transferring the patient into and out of the underwater treadmill is generally easy because patients typically walk into the chamber. The primary concern is safety as well as the comfort of the patient. Therefore, support with a sling may be necessary if the patient is weak or reluctant to enter at first. Weak patients may also benefit from sling support while in the underwater treadmill. This may be supported by the therapist or a support pole that runs through the handle of the sling.

Evaluation in the Underwater Treadmill

After the patient is in the underwater treadmill, the therapist may observe the gait of the patient from the side, front, back or top. Gait evaluation in the underwater treadmill is often very revealing because it is like watching the patient in slow motion, with exaggerated motion of the limbs and joints. If the therapist accompanies the patient in the treadmill, it is possible to palpate joints and muscles while they are moving through voluntary active range of motion. Quality of joint movement, joint integrity, and determination of individual muscle contractions can all be assessed quickly and easily.

Treatment

Techniques for performing rehabilitation in the underwater treadmill include interval training, continuous walking or jogging, walking that is enhanced by passive or manual techniques by the therapist, and walking while wearing a resistive or adaptive device. Interval training is performed early in rehabilitation and conditioning and consists of short periods of slow walking alternating with short rest

periods. This generally results in more rapid progress in strength and cardiovascular conditioning rather than allowing the dog to walk for a longer period until exhaustion. The speed is generally slow to begin with and may be increased every other session if the patient is doing well. The total time may be increased 10 to 15% per week if adequate recovery and progress are made. In enhanced passive or manual walking, the therapist provides manual assistance to limb and joint movement at the appropriate times of the gait cycle to facilitate normal gait or larger range of motion. When doing resisted walking the dog is restrained by holding a vest, or using a leash or harness.

Adaptive devices to enhance underwater treadmill therapy include flotation cuffs, small weight cuffs, and elastic bands. These can be attached to different segments of the limb to encourage exaggerated movement or provide added resistance. Facilitation of joint and limb movement, particularly in attempting to improve active range of motion, is relatively easy in the water.

During the treatment session the therapist must constantly monitor the patient. Be aware of the patient's level of anxiety and fatigue to allow adequate rest periods. Labored breathing or loss of the rhythmic gait pattern are indicators of stress or fatigue. Patients that are older, poorly conditioned, or obese should especially not reach the point of fatigue.

Precautions and Contraindications

There are several precautions and contraindications for underwater treadmill rehabilitation. Patients that have a high level of fear of the water should not undergo aquatic therapy. Typical conditions that do not respond well to underwater treadmill rehabilitation are acute biceps tendonitis and other severe muscle strains as well as fracture repair with external fixators. Aquatic therapy should be used with caution or avoided with cardiac conditions that limit exertion levels, especially in warm water. Patients with borderline respiratory conditions may have difficulty because of the hydrostatic pressure placed on the thorax. Patients with open wounds or surface infections should not be placed in the underwater treadmill. Finally, incontinent patients or patients with diarrhea should not undergo underwater treadmill therapy.

Pool Therapy

Aquatic therapy performed in a swimming pool, or any deep body of water, allows for active range of motion or passive range of motion performed using the warmth, buoyancy, and resistance of water. Pool therapy is used when there is a need for increased muscle or cardiovascular demand and minimal risk of injury or when weight bearing and balance are problems. Secondarily, it may be used to reduce muscle spasm or peripheral edema and improve mood or bolster confidence.¹⁻³

Advantages

Immersion in water to perform exercise has several advantages over an underwater treadmill. The patient is almost completely non-weight bearing and the thoracic and lumbar spines are suspended in a neutral position by water buoyancy. This may reduce the risk of injury and eliminates almost all joint concussive forces. The antigravity muscles are essentially inactive, resulting in only concentric muscle contractions that may minimize the potential for delayed onset muscle soreness. Warm water temperatures increase circulation, caloric expenditure, and nerve conduction velocity; improve coordination; and allow for increased soft-tissue elasticity.¹⁻³ Warming of the tissues through water temperature and increased metabolic demand by way of swimming may also stimulate the release of endorphins and enkephalins that contribute to improving mood, relaxing muscles, and managing pain.³ Additionally, hydrostatic pressure exerted by the water on extremities aids in the reduction of edema.⁴

Pool therapy allows for early intervention leading to faster recovery. It promotes earlier active range of motion of joints especially in flexion, and strengthening of muscles. Functional carryover to land based activities is usually seen very quickly. Another benefit to deep-water therapy is the capacity for performing vigorous sustained exercise especially when land-based activity is not possible, such as in an overweight dog.

Pool Temperature

Most therapy pools are maintained between 82° to 90° F. The most appropriate pool temperature for aquatic rehabilitation will vary depending on the type of facility and the predominant patient population. Outdoor facilities most often keep the water a few degrees warmer in the cool months (88° to 90° F) and a little cooler in the summer (82° to 86° F). The warmer temperatures are most beneficial for relaxation of muscles. The cooler temperatures allow for more vigorous exercise with less risk of overheating. Indoor facilities maintain more consistent temperatures based on their patient population.⁵ When treating a neurologic patient warmer water (85° to 90° F) is preferable. Orthopedic patients and dogs in conditioning programs do not require the same warm temperatures (78° to 86° F). Overheating can be easily avoided by using above-water rests that allow evaporative cooling or by having the dog stand in a small cool pool placed next to the therapy pool. The pool temperatures noted here are variable based on the ambient temperatures if the pool is outside with a range of temperature from 50° to 115° F.

When to Initiate Pool Therapy

Determining when pool therapy should begin is a decision made by the surgeon or veterinarian overseeing patient care. Following most orthopedic surgeries, 14-30 days is

adequate. This allows for sufficient initial tissue healing and initial therapeutic exercises. For surgeries like TPLO, TTA, and joint replacements, 4-6 weeks or more is required to allow early bone healing. Neurologic rehabilitation post-surgery varies depending on the procedure and the quality of the stabilization. Conservative (nonsurgical) neurologic rehabilitation for conditions such as degenerative myelopathy or fibrocartilagenous embolic myelopathy begins as soon as the patient is medically stable, often immediately. Dogs with conservatively managed intervertebral disk disease should not have therapy prior to 1 month after injury to allow a period of healing of the weakened annulus fibrosis. Postsurgical patients with severe spinal cord injuries with minimal motor function may benefit from swim therapy as early as 5 to 10 days after surgery. Particular care should be taken to support the patient and not allow the incision to become wet. These are common time frames for early aquatic intervention and vary from patient to patient.

Equipment

Before getting in the water with the patient, precautions should be taken on the part of the therapist. Nearly every patient is fitted with a flotation vest to provide control and convey a sense of buoyancy to the dog. There are many reliable personal flotation devices (PFDs) made for dogs, such as Cascade Outfitters, Outward Hound, and Ruff Wear. A good PFD conforms well to the body and does not interfere with either shoulder extension or hip flexion. The therapist may want to wear either a shortie or full length wetsuit depending on their skill level and ambient temperature above the water. Wearing a wetsuit minimizes injury by flailing paws or dogs climbing onto the therapist. It is also important to have a supply of float toys, balls, and assistive devices like cuffs made from pool noodles, weight cuffs, elastic bands, rubber bands, and/or hair scrunchies easily accessible. These are used to make therapy fun or encourage different swim strokes.

Transfers

Transferring the patient into and out of the water can be performed in many ways. The primary concern is safety as well as the comfort of the patient. Entering and exiting the water are often the portions of rehabilitation in which the patient is at most risk of injury. Leaping into the water is the least desirable method. When using stairs to enter and exit the pool dogs may push back or push off and this can put an extremity at risk of injury owing to rapid or poorly controlled loading. Steep pool steps also place extremities in more demanding positions. Ramps work well especially for ambulatory patients. Dogs are less likely to launch themselves from a ramp. Stretchers can be most helpful for nonambulatory patients or those with some degree of paralysis. The therapist can easily slide a patient directly from the stretcher into the water. Lifts are often used. They may



Figure 31-7 A harness lift system can also be used.

be hydraulic or electric and use either a sling or a platform to safely place the patient in the pool. A platform lift (see [Figure 31-6](#)) has the added advantage of allowing the patient to be lowered into the pool while its feet remain on a solid surface. Depth adjustability of the platform lift also allows the therapist to control how much weight bearing the patient performs when standing at rest on the lift. This provides an opportunity for balance and weight shift activities during rest periods between swims. A harness lift system can also be used ([Figure 31-7](#)).

Evaluation in the Water

After the patient is in the pool positioning it in front of or beside the therapist allows for evaluation that cannot be performed on land. It is possible to palpate joints and muscles while it is moving through voluntary active range of motion. Quality of joint movement, joint integrity, and determination of individual muscle contractions can all be assessed quickly and easily. As an example, for a hip joint assessment, the therapist places his or her hand flat over the dorsal aspect of the hip while the dog is swimming. A normal healthy hip will move smoothly and all that will be felt is a rhythmic muscle contraction under the evaluator's hand. Changes in articular cartilage or loose bodies in the joint have a crepitant feel, as though a little sand is in the joint. Joint laxity is recognized by a range of sensations from subtle bumping, to clicking, to distinct clunking, to snapping often accompanied by an audible pop. These are indicators of abnormal femoral head movement within the joint capsule during a vigorous swim stroke. Other joints can also be assessed, to a lesser degree, with skillful palpation.

During the evaluation the therapist has ample opportunity to educate the owner about their findings. Encourage owners to ask questions about their dog's condition. In human practice it has been shown there is a strong